Basic Questions

12 questions | Original English questions and source answers

Question 1

What is a function of TFTP in network operations?

Options

A. transfers IOS images from a server to a router for firmware upgrades

B. transfers a configuration files from a server to a router on a congested link

C. transfers files between file systems on a router

D. transfers a backup configuration file from a server to a switch using a username and password

Answer

A. transfers IOS images from a server to a router for firmware upgrades

Source: CCNA 200-301 Questions.html | Question 1

Question 2

What are two functions of a server on a network? (Choose two)

Options

A. achieves redundancy by exclusively using virtual server clustering

B. handles requests from multiple workstations at the same time

C. runs applications that send and retrieve data for workstations that make requests

D. housed solely in a data center that is dedicated to a single client

E. runs the same operating system in order to communicate with other servers

Answer

B. handles requests from multiple workstations at the same time

C. runs applications that send and retrieve data for workstations that make requests

Source: CCNA 200-301 Questions.html | Question 2

Question 3

What is the default behavior of a Layer 2 switch when a frame with an unknown destination MAC address is received?

Options

A. The Layer 2 switch drops the received frame

B. The Layer 2 switch floods packets to all ports except the receiving port in the given VLAN

C. The Layer 2 switch forwards the packet and adds the destination MAC address to its MAC address table

D. The Layer 2 switch sends a copy of a packet to CPU for destination MAC address learning

Answer

B. The Layer 2 switch floods packets to all ports except the receiving port in the given VLAN

Source explanation

If the destination MAC address is not in the CAM table (unknown destination MAC address), the switch sends the frame out all other ports that are in the same VLAN as the received frame. This is called flooding. It does not flood the frame out the same port on which the frame was received.

Source: CCNA 200-301 Questions.html | Question 3

Question 4

What is the same for both copper and fiber interfaces when using SFP modules?

Options

A. They support an inline optical attenuator to enhance signal strength

B. They provide minimal interruption to services by being hot-swappable

C. They accommodate single-mode and multi-mode in a single module

D. They offer reliable bandwidth up to 100 Mbps in half duplex mode

Answer

B. They provide minimal interruption to services by being hot-swappable

Source: CCNA 200-301 Questions.html | Question 4

Question 5

What are two reasons that cause late collisions to increment on an Ethernet interface? (Choose two)

Options

A. when a collision occurs after the 32nd byte of a frame has been transmitted

B. when one side of the connection is configured for half-duplex

C. when the sending device waits 15 seconds before sending the frame again

D. when Carrier Sense Multiple Access/Collision Detection is used

E. when the cable length limits are exceeded

Answer

B. when one side of the connection is configured for half-duplex

E. when the cable length limits are exceeded

Source explanation

A late collision is defined as any collision that occurs after the first 512 bits (or 64th byte) of the frame have been transmitted. The usual possible causes are full-duplex/half-duplex mismatch, exceeded Ethernet cable length limits, or defective hardware such as incorrect cabling, non-compliant number of hubs in the network, or a bad NIC.

Late collisions should never occur in a properly designed Ethernet network. They usually occur when Ethernet cables are too long or when there are too many repeaters in the network.

Reference: https://www.cisco.com/en/US/docs/internetworking/troubleshooting/guide/tr1904.html

Source: CCNA 200-301 Questions.html | Question 5

Question 6

In which way does a spine-and-leaf architecture allow for scalability in a network when additional access ports are required?

Options

A. A spine switch and a leaf switch can be added with redundant connections between them

B. A spine switch can be added with at least 40 GB uplinks

C. A leaf switch can be added with a single connection to a core spine switch

D. A leaf switch can be added with connections to every spine switch

Answer

D. A leaf switch can be added with connections to every spine switch

Source explanation

Spine-leaf architecture is typically deployed as two layers: spines (such as an aggregation layer), and leaves (such as an access layer). Spine-leaf topologies provide high-bandwidth, low-latency, nonblocking server-to-server connectivity.

Leaf (access) switches are what provide devices access to the fabric (the network of spine and leaf switches) and are typically deployed at the top of the rack. Generally, devices connect to the leaf switches. Devices can include servers, Layer 4-7 services (firewalls and load balancers), and WAN or Internet routers. Leaf switches do not connect to other leaf switches. In spine-and-leaf architecture, every leaf should connect to every spine in a full mesh.

Spine (aggregation) switches are used to connect to all leaf switches and are typically deployed at the end or middle of the row. Spine switches do not connect to other spine switches.

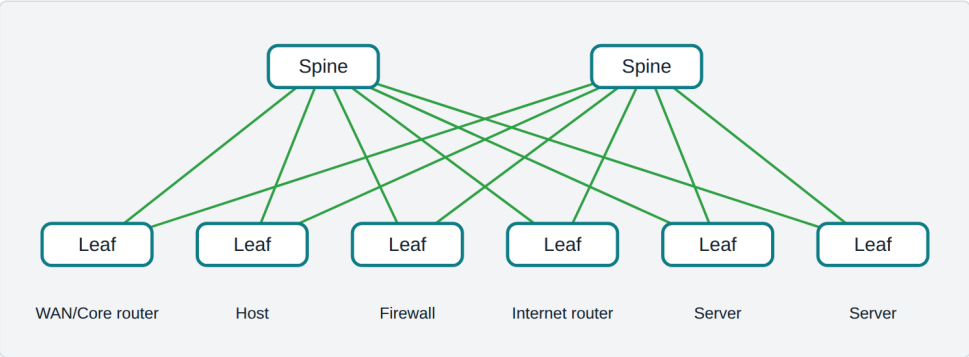
Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-9000-series-switches/guide-c07-733228.html
SpineSpineLeafWAN/Core routerLeafHostLeafFirewallLeafInternet routerLeafServerLeafServer

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Reference: https://www.cisco.com/c/en/us/products/collateral/switches/nexus-9000-series-switches/guide-c07-733228.html



Source: CCNA 200-301 Questions.html | Question 6

Question 7

What is the function of a server?

Options

A. It creates security zones between trusted and untrusted networks

B. It provides shared applications to end users

C. It routes traffic between Layer 3 devices

D. It transmits packets between hosts in the same broadcast domain

Answer

B. It provides shared applications to end users

Source: CCNA 200-301 Questions.html | Question 7

Question 8

Which two statements about the purpose of the OSI model are accurate? (Choose two)

Options

A. Ensures reliable data delivery through its layered approach

B. Defines the network functions that occur at each layer

C. Facilitates an understanding of how information travels throughout a network

D. Changes in one layer do not impact other layer

Answer

B. Defines the network functions that occur at each layer

C. Facilitates an understanding of how information travels throughout a network

Source: CCNA 200-301 Questions.html | Question 8

Question 9

What is the primary function of a Layer 3 device?

Options

A. to transmit wireless traffic between hosts

B. forward traffic within the same broadcast domain

C. to pass traffic between different networks

D. to analyze traffic and drop unauthorized traffic from the Internet

Answer

C. to pass traffic between different networks

Source: CCNA 200-301 Questions.html | Question 9

Question 10

What is the destination MAC address of a broadcast frame?

Options

A. 00:00:0crfHfrff

B. 00:00:0c:07:ac:01

C. ff:ff:ff:ff:ff:ff

D. 43:2e:08:00:00:0c

E. 00:00:0c:43:2e:08

Answer

C. ff:ff:ff:ff:ff:ff

Source: CCNA 200-301 Questions.html | Question 10

Question 11

Which action is taken by a switch port enabled for PoE power classification override?

Options

A. When a powered device begins drawing power from a PoE switch port a syslog message is generated

B. As power usage on a PoE switch port is checked data flow to the connected device is temporarily paused

C. If a switch determines that a device is using less than the minimum configured power it assumes the device has failed and disconnects

D. If a monitored port exceeds the maximum administrative value for power, the port is shutdown and err-disabled

Answer

D. If a monitored port exceeds the maximum administrative value for power, the port is shutdown and err-disabled

Source explanation

PoE monitoring and policing compares the power consumption on ports with the administrative maximum value (either a configured maximum value or the port’s default value). If the power consumption on a monitored port exceeds the administrative maximum value, the following actions occur:

+ A syslog message is issued.
+ The monitored port is shut down and error-disabled.
+ The allocated power is freed.

Source: CCNA 200-301 Questions.html | Question 11

Question 12

A frame that enters a switch fails the Frame Check Sequence. Which two interface counters are incremented? (Choose two)

Options

A. input errors

B. runts

C. giants

D. frame

E. CRC

Answer

A. input errors

E. CRC

Source explanation

Whenever the physical transmission has problems, the receiving device might receive a frame whose bits have changed values. These frames do not pass the error detection logic as implemented in the FCS field in the Ethernet trailer. The receiving device discards the frame and counts it as some kind of input error. Cisco switches list this error as a CRC error. Cyclic redundancy check (CRC) is a term related to how the FCS math detects an error.

The “input errors” includes runts, giants, no buffer, CRC, frame, overrun, and ignored counts.

The output below show the interface counters with the “show interface s0/0/0” command:

```
Router#show interface s0/0/0
Serial0/0/0 is up, line protocol is up
Hardware is M4T
Description: Link to R2
Internet address is 10.1.1.1/30
MTU 1500 bytes, BW 1544 Kbit, DLY 20000 usec,
reliability 255/255, txload 1/255, rxload 1/255
--output omitted--
5 minute output rate 0 bits/sec, 0 packets/sec
268 packets input, 24889 bytes, 0 no buffer
Received 0 broadcasts, 0 runts, 0 giants, 0 throttles
0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored, 0 abort
251 packets output, 23498 bytes, 0 underruns
0 output errors, 0 collisions, 0 interface resets
0 output buffer failures, 0 output buffers swapped out
0 carrier transitions DCD=up DSR=up DTR=up RTS=up CTS=up
```

Source: CCNA 200-301 Questions.html | Question 12